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A Leap Towards World Class Education

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Name of the Experiment:

Performed on:
Submitted on:

Marks	Teacher's Signature with date
10	

Que 1 List the python sequence data types and state one key characteristic of each with respect to mutability or ordering

Python has several sequence data types that store ordered collections of elements.

1) List : Ordered and Mutable (element can be changed).
Example: [1, 2, 3]

2) ~~Tuple~~ Tuple - Ordered but immutable (cannot be modified)
Example : (1, 2, 3)

3) String - ordered but immutable (sequence of characters and immutable.
Example : "Python"

4) Range : Represents a sequence of numbers and is immutable. Example : range (1,10)

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Que: Explain how indexing and slicing work in python sequences. Illustrate how they differ when applied to string and lists.

⇒ Indexing is used to access a single element from a sequence using its position.

Index starts from 0.

Example:

```
S = "Python"
Print (S[0])        # P
Print (S[3])        # h
```

Slicing is used to access a range of elements from a sequence.

Example:

```
S = "Python"
Print (S[1:4]       # yth
```

Difference in Strings and lists.

- Strings are immutable
- Lists are mutable

```
List 1 = [10, 20, 30]
List1[1] = 50          # allowed.

S = "Hello"
# S[1] = "a"           # not allowed.
```

Que 3 Write a program that:
→ uses indexing and slicing on a string or list, and
→ Applied at least two built-in functions to display result.

Ans 3) Python program using indexing and slicing.

```
name = "Programming"

# indexing

Print ("first letter:", name[0])
Print ("Last letter:", name[-1])

# slicing

Print ("first 5 letters:", name[0:5])
Print ("Last 4 letters:", name[-4:])

# Built-in function.

Print ("Length of string:", len(name))
Print ("Maximum character:", max(name))
```

Que 4 Design and implement a python program that uses.
→ Appropriate sequence types
→ Indexing or slicing
→ Build-in Function.

```
Students = ["Rahul", "Aman", "Neha", "Riya"]
# Indexing
Print ("first student:", Students[0])
```

```
# Slicing
Print ("first two students :", Students[0:2])

# Build-in function.
Print ("Total students :", len(students))

# Add new student
Students. append ("Karan")
Print("Updated list :", Student)
```

Qu 5 Evaluate the suitability of lists, sets, and dictionary for storing and managing unique student records Justify which data structure is most efficient and why.

→ List – stores ordered data but allows duplicate values

Set – Stores unique elements only, but dosnot maintain order

Dictionary – Stores data in key-value pairs.

Example :-

Student = {"ID" : 101, "Name" : "Rahul", "Marks" : 85}

Dictionary is most efficient for student records Because data can be accessed quickly using keys like ID or Name

Que 6 Design and implement a Python program that uses: Appropriate sequence types, Indexing or slicing, Built-In functions:

In Python, (==) and ~~checks~~ is ~~whether~~ are comparision operators.

- Equality (==) checks whether two variables have the same value
- Identity (is) checks whether two variables refer ~~to~~ the same object in memory

Example:

```
a = [1, 2, 3]
b = [1, 2, 3]

Print (a == b)    # true (values are same)
Print (a is b)    # false (different memory location)
```